Ranger: A Ground-facing Camera-based Localization System for Ground Vehicles

Kristopher Kozak
Southwest Research Institute
San Antonio, Texas 78238

Marc Alban
Southwest Research Institute
San Antonio, Texas 78238

***Abstract*—This paper presents an overview of the Ranger localization system and its constituent technologies and algorithms. Ranger is a high-precision localization system for ground vehicles that performs map-based localization using a ground-facing camera. Ranger uses commercially available hardware, including a camera, lights, and a computer, in combination with auxiliary localization sensors and a custom state estimator, to produce a complete high-precision positioning solution suitable for feedback control of an automated vehicle. Ranger was originally conceived and designed to address the accuracy and availability problems of GPS, and can operate independent from or as a supplement to GPS and other GNSS positioning systems. Ranger position measurements are made by matching live ground imagery to imagery stored in a map. The image matching process uses a feature-based approach that yields a high positive match rate with a vanishingly small false positive rate. The precision of the Ranger measurements is shown to be on the order of centimeters, and the actual lateral positioning performance of MARTI, one of Southwest Research Institute's automated vehicles, autonomously driving a test route is shown to be repeatable over many runs to within 2 cm.**

I. Introduction

Ranger is a form of map-based localization that provides very high-precision measurements with or without GPS. Map-based localization has become a popular solution to the accuracy and availability problems of GPS, and map-based localization schemes exist for various sensor modalities, with LIDAR- [1] and camera-based [2] methods being common for on-road automated vehicles. The Ranger approach utilizes a ground-facing camera, lights, and a computer. A notional schematic of a Ranger system is shown in Figure 1. Ranger pairs a single ground-facing camera with synchronized illuminators mounted underneath a vehicle to capture well-illuminated images of the ground that are free from motion blur at vehicle speeds up to 130 km/h. Ranger localization is based on the idea that a patch of ground, almost regardless of how nondescript to the eye, is uniquely identifiable based on the appearance and distribution of visual features on that patch.

For most types of pavements (asphalt and concrete, for instance), the mathematical image features frequently correspond to distinct visible features such as stains, cracks, tar patches, and exposed aggregate, but other less-distinct features and textures can also uniquely distinguish ground patches. The uniqueness of patches of ground serves as the basis for Ranger localization—matching a current view of the ground to a previous observation of the same patch of ground provides very precise position information. Feature-based image matching is a well-studied concept, and Ranger exploits many of the associated techniques to ensure a high rate of valid matches while also preventing the occurrence of essentially all false positives in the process.

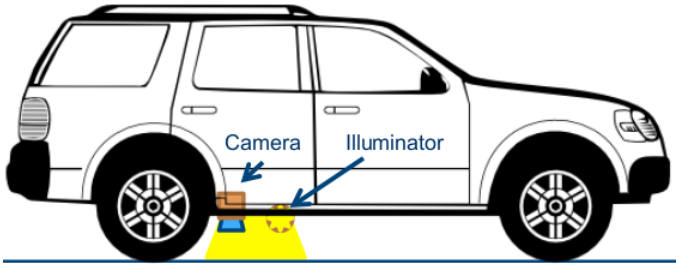


Fig. 1. Simple schematic of Ranger: a camera, lights, and a computer.

Ranger has been under development since 2011, and over that time, the system has been installed on several different kinds of vehicles, including three owned by Southwest Research Institute (SwRI). Figure 2 shows the SwRI vehicles on which Ranger is installed, along with a view of how Ranger is mounted to the undercarriage of each vehicle. The vehicle on the left is an automated HMMWV that served as the original development platform for Ranger, chosen because of its generous ground clearance, high-capacity electrical system, and substantial computing resources. The vehicle in the center is a 2014 Acura RLX that has been used for testing Ranger on freeways. The vehicle on the right is an automated 2006 Ford Explorer named MARTI on which the bulk of the most recent development and testing has been carried out, including the testing reported in this paper. Ranger has been successfully operated and tested on a variety of road surfaces (new asphalt, weathered asphalt, concrete, etc.), in dry and rainy conditions, and on freeways at high speeds.

The purpose of this paper is twofold. First, it provides an overview of Ranger and its various constituent technologies and algorithms. Second, it presents experimental results that reflect various aspects of Ranger performance, including the precision of measurements and the suppression of false positive image matches. In these experiments, performance of the Ranger system was evaluated under nominal, benign

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Fig. 2. SwRI has installed Ranger on a variety of vehicles. Left: SwRI's automated HMMWV. Center: 2014 Acura RLX. Right: MARTI—A 2006 Ford Explorer, SwRI's first autonomous vehicle.

conditions and at relatively low speeds.

II. System Design

The Ranger system consists of three main components: a camera, illuminators, and a computer. The process of imaging the ground beneath a vehicle in a consistent manner presents a number of challenges. Among the most significant are minimizing motion blur, especially when traveling at high speeds, and ensuring adequate illumination and exposure, especially in low light conditions. Ranger addresses motion blur by simply capturing images with a very short exposure. With the camera pointing directly at the ground, the maximum exposure time to prevent motion blur is based on the speed of the vehicle and the size the ground region imaged by each pixel. This design assumes a top speed of 130 km/h (approximately 80 mph), a practical upper limit for passenger vehicles. The size of the ground region imaged by each pixel, or pixel scale, is determined by the camera resolution and field of view (but note that the images can be resized to yield an arbitrary pixel scale). Ranger systems, including the one used to generate the experimental results for this paper, have generally used a pixel scale of 1.5 mm, which requires an exposure of shorter than 40 µs to prevent motion blur.

With a typical camera, exposure times that short prevent resolving images in ambient lighting conditions other than direct sunlight. As a result, artificial illumination is a necessity. The estimated amount of light needed to adequately resolve an image is far too high to be practical with traditional light sources (*e.g.*, incandescent and fluorescent). Approximately forty 100 W incandescent lights would be necessary to provide the light levels needed for Ranger. Such a lighting system, if realized, would be both intensely bright and impractical to power from passenger vehicle electrical systems. Fortunately, by the start of this project, high-intensity LED arrays, designed for commercial lighting applications, had become widely available. In addition to consuming far less power per lumen, these LED arrays are significantly smaller than equivalent traditional lighting technologies and can be pulsed at a high rate to ensure full light output during the camera exposure while minimizing the average power draw and the apparent brightness.

A. Camera

One of the goals of this project was to use commercially available, and inexpensive, hardware when possible. For cameras, this goal was easily met. Ranger has benefited substantially from the many recent advances in camera technology. Various characteristics of camera performance, such as sensitivity, resolution, and frame rate, have improved significantly, and costs have fallen. As a result, there are currently a number of suitable camera options for Ranger. In addition to the camera, special consideration was made for the optics. Mounting the camera underneath a vehicle means the height from ground is inherently limited. To image a large enough patch of ground to be practical, a wide-angle lens is typically necessary. To avoid adding an image rectification step to the image processing pipeline, low-distortion lenses have been preferred over fisheye-type lenses. Although low-distortion lenses are longer (and more expensive) than equivalent fisheye lenses, integration of the camera enclosures has not yet been an issue. Moreover, for smaller vehicles, the field of view has tended to be limited by occlusions from the undercarriage rather than available height above the ground.

The actual fields of view (referenced to the rectangular patch of ground imaged by each camera) for the vehicles shown in Figure 2 are 1.4 meters wide by 1.0 meters long for the HMMWV, 1.0 meter wide by 1.2 meters long for the Acura RLX, and 2.0 meters wide by 1.5 meters long for MARTI. The camera on the HMMWV (the original Ranger prototype) was different from subsequent installations, which is why the field of view was smaller than that for MARTI despite the camera being mounted higher.

Operation of the camera is straightforward. Image frames are triggered via an external signal from a microcontroller, which also triggers the illuminators. The exposure time for each frame is set in software and is generally held fixed or adjusted online through a separate software process rather than with the camera's built-in auto-exposure function. The external frame trigger operates at 60 Hz, the maximum frame rate of the camera; however, the frame rate can be subsequently throttled in software to reduce processing load associated with image transport and acquisition. When collecting images for mapping, the camera is allowed to run at full frame rate to ensure good coverage of the ground; when localizing, the frame rate is throttled to approximately 10 Hz.

B. Illuminators

At the outset of this project, the use of near-infrared (NIR) illuminators, which have the benefit of producing no visible light, was investigated; however, although image sensors with good NIR sensitivity have become readily available, sufficiently bright NIR LEDs have not. High-intensity, visible (white) light LED arrays, on the other hand, have recently proliferated and provide an economical option for producing the extremely high levels of light needed to resolve ground images. Through the course of Ranger development, the amount of light required for robust matching has been gradually minimized—usable data can be extracted from seemingly

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black images (see Figure 3, for example). Nevertheless, a substantial amount of light is still necessary to resolve images with low visual noise. Luminous flux levels of 40,000–70,000 lumens have been empirically determined to be sufficient, given the sensitivity of the camera sensors and the aperture of the optics. Manageable quantities of small-form-factor LED arrays are capable of providing this level of light output. For instance, MARTI uses twelve 5,600-lumen LED arrays to generate a total of 67,200 lumens. The instantaneous power draw of these lights is approximately 600 W (50 W per array).

The LEDs are powered from a relatively small DC-DC power supply that steps up standard vehicle voltages to the levels required by the LED arrays. To accommodate the high instantaneous power draw of the lights, and to avoid the need for an oversized power supply, a suitably sized capacitor was incorporated into the lighting circuit to supply the high currents needed to to drive the lights while on and recharge when the lights are off. The LEDs are pulsed at 120 Hz to avoid visible flicker. The LEDs are turned on slightly in advance of the camera trigger to ensure that full brightness is achieved before the camera begins capturing an image, and the LED on-time, typically 120 µs, is set to accommodate the longest expected camera exposure. These values result in a duty cycle of 1.44%, which, for MARTI, results in an average power draw of less than 10 W.

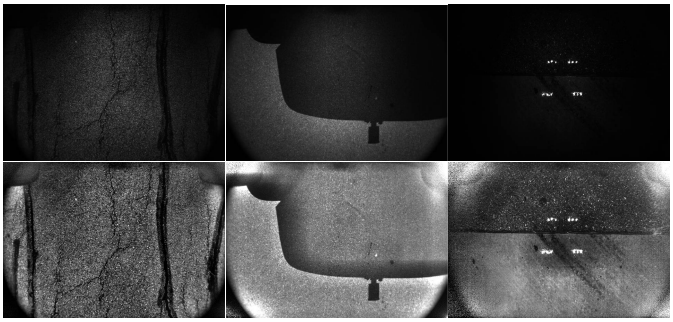
The initial illumination system design for the first Ranger prototype focused on providing highly uniform lighting. To that end an illumination simulator was developed to test various configurations of lights and allow for placement studies to identify the best mounting locations on vehicle undercarriages. The simulator incorporated the specific light distributions of the LEDs (and reflectors, when used) and also included vignetting as measured empirically. While this was a useful tool to understand and try different configurations, it has generally been more efficient to simply install the lights in a nominal configuration and then adjust the lights (move, add, or remove) based on the actual light distribution. This process is sufficient because the uniformity of the lighting has little effect on matching performance. Image noise, on the other hand, which is a symptom of light levels that are too low, significantly degrades image matching performance. Therefore, the best arrangement of the lights is less a matter of uniformity and more one of coverage.

C. Computing

Ranger was developed to operate on consumer-grade desktop CPUs. Beyond this, no special computing hardware is required. For reference, the majority of the development for Ranger, and the experiments carried out for this paper, were performed on a computer with an Intel Core i7-4770T CPU and 16 GB of RAM.

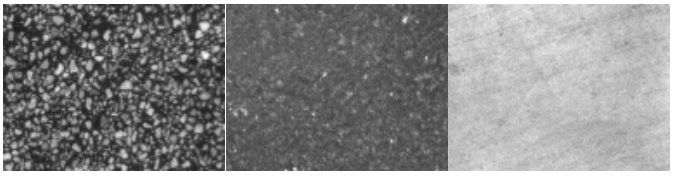
III. Image Matching

The core of the Ranger approach is accurate and robust matching of images between the current camera view and an existing map of images. Like other existing image-based localization approaches [3] [4], Ranger relies on distinctive local image descriptors for this task. The positioning of the Ranger camera limits the scene to effectively a planar surface at a constant scale and distance, which makes the use of conventional geometrically constrained correspondence techniques [5] very efficient and robust.



(a) nominal (b) direct sun (c) wet

Fig. 3. Contrast stretching for different surfaces and lighting conditions. Top: without contrast stretching. Bottom: after applying contrast stretching.



(a) worn asphalt (b) dark asphalt (c) concrete

Fig. 4. Close-up images of three different road surfaces as captured by the Ranger camera.

A. Contrast Stretching

Before further processing the images, it is first necessary to normalize and enhance the contrast of the images. This adjustment ensures that salient features are detected uniformly across the entire image and not just from the areas of highest contrast in the unnormalized image. Ideally, the illumination should be very uniform, but in practice it is often very non-uniform. Moreover, ambient sunlight with shadows in the frame and specular reflections from the LEDs tend to exacerbate non-uniformity. To minimize the impact of this variation, a windowed dynamic contrast stretching method was developed to normalize the illumination and enhance the contrast across the image regardless of the instantaneous light distribution. The approach divides the image into a 16 x 16 grid and calculates the minimum and maximum intensity values of each cell. Each pixel intensity is then scaled based on the minimum and maximum intensity interpolated from the 4 nearest grid points. Contrast stretching artifacts from specular reflections are minimized by first masking out areas that are over-exposed. Figure 3 shows the effect of contrast stretching on a small sample of different surface and lighting conditions.

B. Feature Detection and Description

Feature detection is a critical step in the matching process. It is necessary that a subset of the detected features in

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two images of the same subject actually correspond to the same physical points. For Ranger matching to be robust, this must remain true in the presence of different translations, rotations, illuminations, and environmental conditions. Due to the constrained field of view of the camera, scale invariance is not a significant concern.

Next in importance is the description of the keypoints, which must represent the features in a space where the distance between physically corresponding features between two images are closer than non-corresponding features regardless of their locations in the images. Again this must remain true in the presence of different translations, rotations, illuminations, and environmental conditions.

TABLE I
DETECTOR/DESCRIPTOR PAIR PERFORMANCE NORMALIZED TO BEST PERFORMACE FOR EACH SURFACE

	worn asphalt	dark asphalt	concrete
CenSurE / SIFT	1.0	1.0	1.0
CenSurE / ORB	0.96	0.81	0.99
SIFT / SIFT	0.9	0.87	0.95
CenSurE / BRISK	0.92	0.64	0.99
CenSurE / FREAK	0.83	0.31	0.87
FAST / SIFT	0.92	0.99	0.11
FAST / ORB	0.91	0.73	0.21
FAST / BRISK	0.91	0.58	0.1
SURF / SURF	0.78	0.38	0.39
GFTT / SIFT	0.90	0.49	0.08
GFTT / ORB	0.91	0.44	0.09
GFTT / BRISK	0.89	0.39	0.06
FAST / FREAK	0.78	0.27	0.07
GFTT / FREAK	0.66	0.13	0.01
CenSurE / SURF	0.01	0.01	0.41
FAST / SURF	0.06	0.0	0.0
GFTT / SURF	0.0	0.0	0.0

The best overall results were achieved using the CenSurE detector [6] and ORB descriptor [7]. Over the course of development we also evaluated the performance of SIFT [8], SURF [9], GFTT [10], FREAK [11], BRISK [12], and FAST [13] detectors and descriptors. The SIFT descriptor actually performs slightly better than ORB, but it is significantly more expensive to compute and requires a license for commercial use, limiting its usefulness for our applications.

Table I shows the relative performance of each feature detector and descriptor pair compared to the best-performing configuration on different common road surfaces. The performance was measured as the number of successfully matched pairs of partially overlapping images from a dataset of thousands of images. Characteristic examples of the worn asphalt, dark asphalt, and concrete road surfaces used for this measurement are shown in Figure 4. In these data sets, the CenSurE detector consistently outperforms the other approaches and the CenSurE / SIFT configuration gives the best performance, followed closely by CenSurE / ORB.

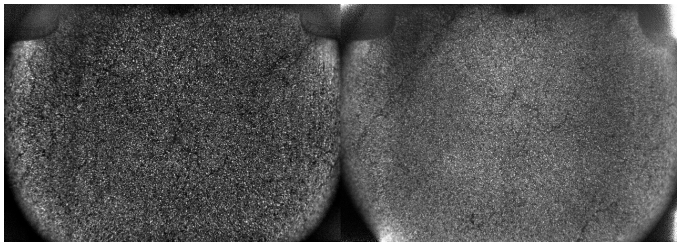


Fig. 5. Overlapping images of asphalt patch.

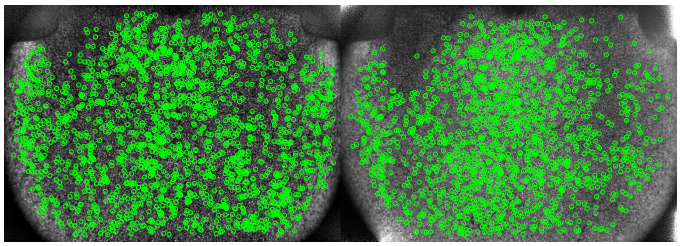


Fig. 6. Keypoints detected in both images

C. Feature Correspondence

Initial correspondences between two paired images a and b are computed by comparing each feature in image b to each feature in image a to find the closest matching features. In the case of binary descriptors, such as ORB, Hamming distance [14] is used for the comparison. Cross-checking is performed such that correspondences are returned only if the closest matching descriptor is consistent in both directions (from a to b and b to a), which has the effect of excluding some of the invalid matches.

By themselves, the keypoint correspondences generated from descriptor matching are weak and unreliable indicators that two images match. Figure 7 shows the correspondences from the raw descriptor matching step, which appear random. Even when two images significant overlap one another, as is the case here, between 70 and 90 percent of the correspondences may be false. Furthermore, images that don't overlap will have a similar total number of feature correspondences. This is understandable since at the local level of the features, a given location of pavement will be similar to many others. Asphalt and rigid pavements are fairly uniform in composition even if the distribution and texture is somewhat randomized.

Due to the constraints of the camera position, a set of true correspondences between images will approximate a rigid 2-D transform. RANSAC [15] is used to efficiently sample and fit a rigid transform to a set of correspondences. This approach benefits heavily from the simplicity of the model, which requires only two correspondences per sample, compared with a 2-D homography, which requires four, and the fundamental matrix for epipolar geometry, which requires eight. In the presence of such a large percentage of outliers, a simpler model greatly increases the probability that a valid set of inliers will be selected for the hypothesis within a reasonable number of iterations.

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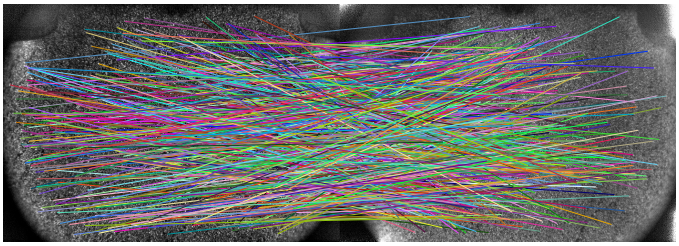


Fig. 7. Brute force feature correspondences.

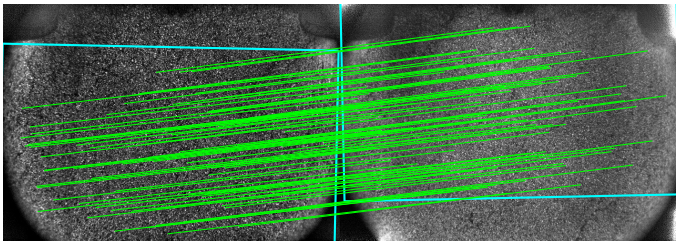


Fig. 8. Feature correspondences consistent with a rigid transform.

To account for the fact that the rigid 2-D transform is only approximate—for example, pitch and roll of the vehicle induce perspective distortion in the image, which violates rigid transform assumptions—the reprojection error threshold for the RANSAC fit is left very loose. The initial fit is then followed by a 4-point homography fit using RANSAC to catch and discard any false matches that made it through the first step. The high percentage of inliers in the sample after the rigid fit results in this step being extremely fast. Finally, a least-squares regression for the 2-D rigid transformation is calculated from the remaining correspondences. This transformation, which connects the live image to its matching image in the map, is then used to identify the position of the camera in the world coordinate system.

IV. Map Creation

Another important part of the Ranger system is the creation of the map used to facilitate localization. A detailed discussion of the mapping process is beyond the scope of this paper, but a brief overview of the approach and the key elements is provided here for context. Ranger maps are created from and consist of strips of sequential overlapping images. These strips of images are recorded by the mapping Ranger system (which may be the same as the localizing Ranger system) along with global position from GPS, yaw rate, and vehicle velocity data, which are acquired from auxiliary localization sensors. Typically, one strip is collected for each lane of travel. The map images are processed in a manner identical to live localization images to enhance contrast, detect keypoints, and extract feature descriptors. The image feature data is then stored to the map along with registration data that defines the location of each frame in a world coordinate system. The feature data is then used directly in the image matching process, while the map images themselves are retained only for development, debugging, and visualization purposes.

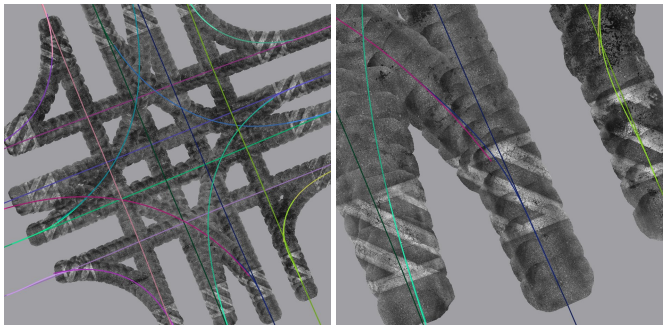


Fig. 9. Ranger map of a complex intersection after optimization. The ground images for each map frame are rendered to provide a visual indication of the registration accuracy after optimization.

To ensure consistency when a map is composed of multiple segments that overlap one another, the position and orientation in the world coordinate system of each image from the map is refined in an optimization step. The optimization balances factors such as the original measured position of the camera (captured during the map data collection process from a GNSS receiver) with the relative position of sequential frames based on the kinematics of the map data collection vehicle using relative position constraints between overlapping image frames (*e.g.*, single-segment loop closures and inter-segment overlaps). These loop closures and other non-sequential overlapping frames, which occur at locations such as intersections, are detected using the image matching approach described in the previous section. The optimization is formulated as a nonlinear least-squares optimization problem and is solved using Ceres Solver [16].

While the optimization can improve the registration accuracy of the map as a whole, the primary reason for performing the optimization is to ensure consistency between segments. For complex maps with many overlapping segments, registration inconsistencies can manifest as localization errors. The map optimization approach developed for Ranger has generally succeeded at ensuring local consistency between overlapping frames while also not distorting other parts of the map. Figure 9 shows an example of a portion of a complex Ranger map after optimization.

V. Localization

A direct approach is taken to search the map for potential matches to the current view of the ground. The set of potential matching map images is sorted based on distance from the current estimated position of the vehicle. Each map image is then compared with the current image until a match is found or the search times out. An image is considered to match if at least 25 feature correspondences consistent with a rigid transform are found. GPS is sufficient to initialize the vehicle position, and after the first match, subsequent matches are usually found with only one comparison. On the live system, four processing threads are devoted to handling incoming images, supporting an output rate of 10 Hz.

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Ranger utilizes a variant of an Extended Kalman Filter (EKF) to provide a continuous and filtered estimate of the pose and velocity of the vehicle. In contrast to a standard EKF, the Ranger EKF was designed to accommodate measurements that exhibit high and variable latency characteristics. The overall processing time for a single Ranger measurement averages around 100 ms per measurement. Moreover, the maximum processing time is bounded by only the user-configurable timeout on the search—as the number of map frames searched increases, so does the processing time. For the Ranger measurements to be usable for practical applications, particularly for feedback control, the EKF must accommodate these latency characteristics.

The latency of a Ranger measurement is affected by two main factors: the image pre-processing time and the image matching time. Image pre-processing operations occur for every new frame and include image acquisition, rectification, scaling, and contrast enhancement, as well as feature detection and extraction. Because these pre-processing operations happen only once per frame, their contribution to latency is bounded. The largest contributors to latency for this step are the feature detection and extraction steps. For the Ranger system on MARTI, feature detection typically takes 30 ms per frame for the CenSurE detector, though it can take over 100 ms for some frames when the detector must be run multiple times to yield a target number of features. Feature extraction for the ORB descriptor consistently takes approximately 20 ms per frame. The contribution of the image matching operation to latency is consistent on a per-frame basis, but as an iterative search, the total latency can grow as more frames are searched. For each frame the brute-force matching operation typically takes about 30 ms while the RANSAC step takes about 10 ms. The sum of these latencies gives a minimum Ranger latency time of 90 ms.

Like a traditional EKF, the Ranger EKF fuses position and heading measurements with measurements from other available localization sensors. Auxiliary localization sensors include IMU, wheel speed sensors and GPS. The EKF uses an Ackerman steered vehicle model with the angular rate and longitudinal velocity measurements for motion prediction. EKF updates occur at a fixed rate (typically 50 Hz). To handle the high latency of Ranger measurements, a rolling history of states and measurements is retained. Each filter update step is then carried out from the oldest unprocessed measurement rather than from the time of the last update. In other words, when high-latency (old) data is received, the entire state history from the time stamp of that old measurement is recomputed. A similar approach was described for a sigma-point Kalman filter by Ziegler, et al. [2]. In addition to providing smooth, high-rate state estimates for control, the output of the EKF provides a seed location for the Ranger image matching process to start.

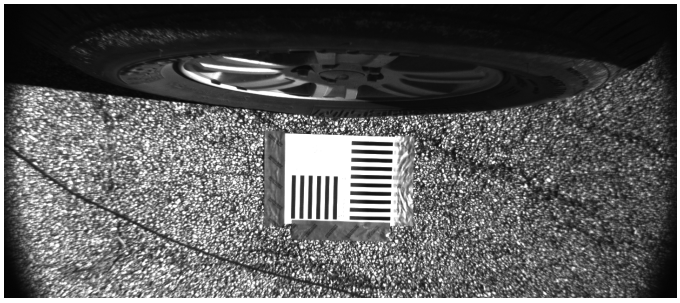


Fig. 10. Image from the external measurement camera of one of ten scale tags used to measure lateral offset of vehicle over multiple runs. The black stripes on the tag are each 1 cm wide and have a pitch of 2 cm.

VI. Experimental Results

A. Precision

For this paper, the assessed characteristic of Ranger localization performance is the lateral precision of the position measurements. Lateral precision represents the run-to-run variation of measurements along the lateral axis of the vehicle, assuming the vehicle follows the same path for each run. Although longitudinal precision is also important in many contexts, measuring it accurately is challenging under dynamic conditions. This is primarily because timing uncertainties in the evaluation of precision are difficult to distinguish from actual longitudinal error. Furthermore, requirements on lateral positioning precision for automated vehicle applications tend to be more stringent and constant across a wide range of speeds, while longitudinal precision requirements can vary significantly depending on the specific application (for example, parallel parking a vehicle requires tight control of longitudinal position, while lane keeping on a freeway does not).

Evaluating the precision of Ranger measurements requires an apparatus or a secondary measurement system that can provide ground truth measurements with a precision greater than that of Ranger. Several methods have previously been employed to measure precision accurately, including the use of a high-accuracy GPS unit with RTK corrections and hand measurements of vehicle offset from fixed reference points; however, these approaches generally did not provide sufficient precision themselves (or in the case of the hand measurements, the ability to make measurements while the vehicle was in motion) to accurately evaluate Ranger performance.

The Ranger system on MARTI (Figure 2) was used for this experiment. To measure the lateral precision of the Ranger measurements, MARTI was allowed to autonomously drive a fixed route that coincided with a segment of the Ranger map at the SwRI test track. Ranger measurements filtered by the EKF were used directly for the feedback control. The specific Ranger map that was used for this test was created in May 2015, while this test was carried out in January 2016. The section of the course used for evaluation of Ranger measurement precision was an approximately 70-meter curving stretch, shown in Figure 11. Measurements were taken at ten chosen locations along the evaluation section. Vehicle

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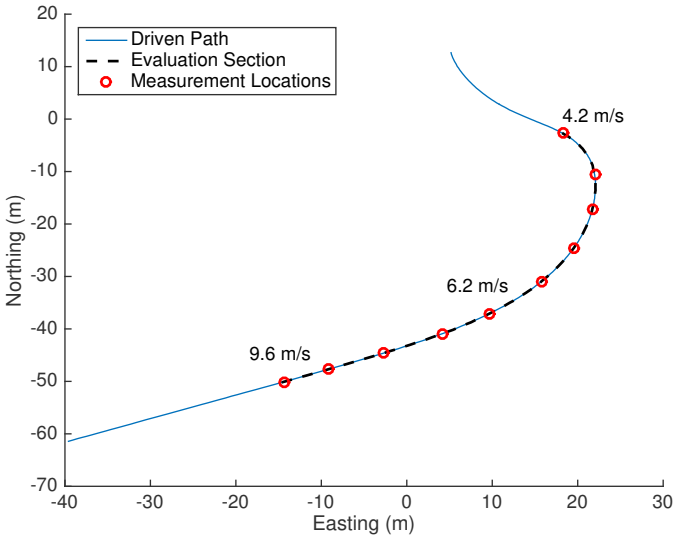


Fig. 11. A portion of the driven path with the evaluation section highlighted. Circles indicate the locations of individual measurements; the beginning, middle, and end measurement locations also indicate speed at that location.

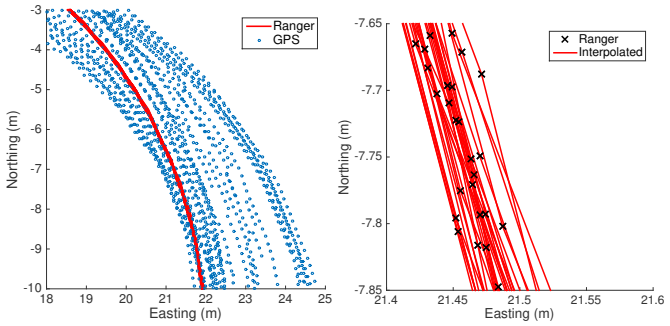


Fig. 12. Left: Close-up of a sample of Ranger measurements (solid red line) in comparison to GPS (blue dots) over 32 automated passes of the test course. Right: Close-up of Ranger measurements.

speeds ranged from 9.6 m/s at the start of the evaluation section to 4.2 m/s at the end.

To ensure reliable measurements of lateral offset at the measurement locations along the path, a second camera was mounted above the driver side rear wheel. This camera was used to measure the lateral offset of the vehicle, or, more specifically, the offset of the rear tire relative to a fixed point on the ground, at each measurement location for each pass. Scale tags were affixed to the pavement at each of these locations, as shown in Figure 10, to provide the fixed point of reference on the ground and an accurate scale to facilitate the conversion from pixels to centimeters. The actual scale of the images from which the measurements were made was approximately 4.5 pixels per centimeter.

A total of 25 runs were carried out for the test. Figure 12 shows a close-up comparison of Ranger measurements to GPS measurements captured during these 25 test runs and 7 pre-test runs. Images and raw data produced by the Ranger system and external camera were collected for each run. The images

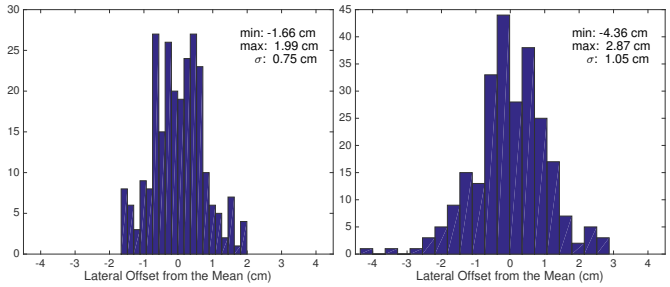


Fig. 13. The distribution of measurements of lateral offset aggregated from all 10 measurement points. Left: Lateral offset measurements obtained from the externally mounted camera, Right: Lateral offsets as measured by Ranger after subtracting the corresponding path offset

containing the scale tags were extracted from recorded data automatically, while the measurements of lateral offset (the distance converted from pixels between the tire and the scale tag) were made manually from each image. The precision of these manual measurements was estimated to be on the order of 2–3 pixels or 0.44–0.67 cm. In Figure 13, the left image shows the distribution of lateral offset measurements for all ten of the measurement locations as obtained from this manual process. This particular set of measurements does not indicate Ranger precision directly, but it is clear that with a total lateral variation of less than 4 cm, Ranger *enables* very precise path following. The right image in Figure 13 shows the distribution of lateral offsets, relative to the mean, as measured by Ranger. This measurement does not capture precision exactly, because the actual path offset contributes to the variance.

To assess the measurement precision, the manually-measured path offset was subtracted from the corresponding Ranger-measured offset. After this correction, the standard deviation of the distribution falls from 1.05 cm to 0.87 cm. The minimum and maximum values remain similar at -3.74 cm and 3.07 cm, respectively. While this precision is quite good, it is somewhat surprising that the actual lateral offset of the vehicle was significantly less than the offsets measured by Ranger, considering that Ranger was the source of the measurements for feedback control of the vehicle. It is possible that the the lower observed variance of the path can be attributed to the filtering effect of the EKF. It is also possible that the uncertainty in the locations of the scale tags with respect to the Ranger maps contributed to the observed variance. The actual tag locations were unfortunately not measured directly and could therefore only be estimated after the fact.

B. False Positives

A significant fundamental assumption of the Ranger localization approach is that the likelihood of mismatching a live image to the wrong map frame is negligibly small. To verify this assumption, Ranger was tested with over 1 million non-overlapping image pairs to characterize the ability of the approach to filter out false positives. The number of inlier feature correspondences, those that are consistent with a rigid 2-D transformation, for each tested image pair were calculated

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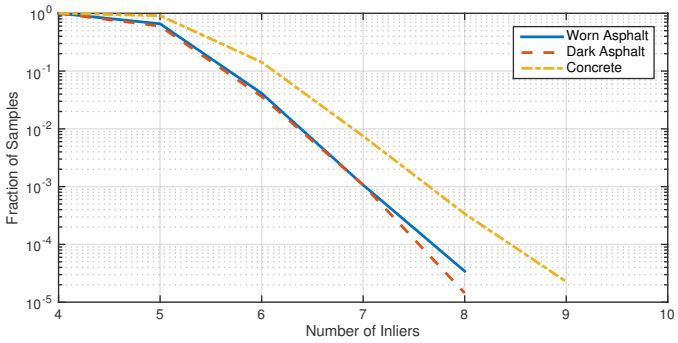


Fig. 14. False positive analysis showing fraction of non-overlapping image pairs containing a given number of feature correspondences consistent with a rigid transform for over 1 million image comparisons on various road surface types. The fraction is shown on a log scale.

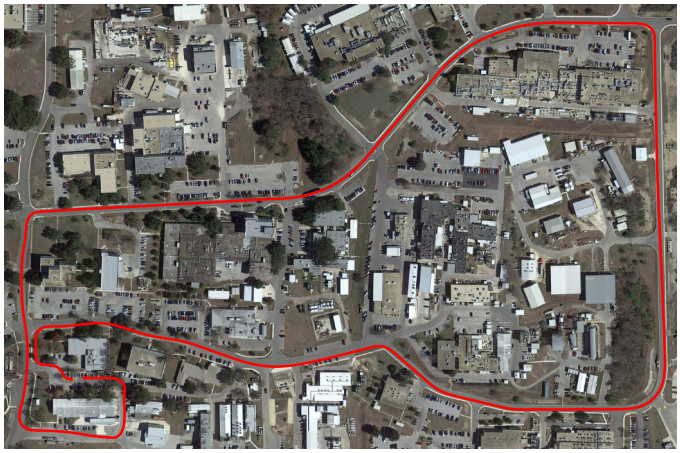


Fig. 15. Autonomous course driven by MARTI over 2.5 km of SwRI roads using only RANGER for localization.

to determine the likelihood that a potential match with a given number of inliers is actually a false positive.

Figure 14 shows the fraction of non-overlapping image pairs containing at least a given number of inliers. This analysis was split by surface type and performed for dark asphalt, worn asphalt, and concrete. In each case no false matches were returned with more than 9 inliers. This is a very strong result with respect to the number of inliers of actual matches which is usually between 100 and 200, while the standard threshold used for a valid match is 24 inliers.

This is evidence that highly repetitive structures are not common at this scale in these common road surfaces. This is a benefit over more conventional image matching approaches where repetitive structures, such as building facades and signs may result in much more frequent false correspondences.

C. Live Performance

Finally, while precision and suppression of false positives is very important, perhaps even more important is the ability of Ranger to consistently and reliably detect matches while on a map, and to do so under the wide range of normal conditions encountered by passenger vehicles. In live autonomous testing, the Ranger system is able to produce matches at a rate of 10 Hz with effectively no drop-outs in the tested environments. This performance is sufficient, along with state estimation, for reliable and repeatable autonomous control to within centimeters of the desired path. Figure 15 shows the autonomously driven path of the MARTI vehicle along 2.5 km of SwRI roads using only Ranger for localization.

VII. Conclusions and Future Work

This paper has provided an overview of the composition, function, and performance of SwRI's Ranger high-precision camera-based localization system for ground vehicles. SwRI's MARTI autonomous vehicle was used to measure the performance of Ranger on a test course, from which the lateral positioning precision of Ranger was measured to be better than 1 cm (at a 1-σ level). Moreover, the actual observed lateral offsets from run to run under autonomous control were less than 2 cm from the mean over the entire course.

At the outset of this project, there were many questions about the feasibility of a system like Ranger, but after four years of active development, most of those questions have been addressed, and the feasibility of Ranger as a practical localization system has been demonstrably proven. Nevertheless, there are a number of practical issues that remain. Many of the interesting questions relate to the persistence and stability of maps over time. For example, while it is clear that ground imagery is relatively stable over fairly long timescales, there has been no thorough investigation into how gradual changes to the road surface degrade the image matching and localization performance. Ranger has performed successfully on maps that were built from ground imagery that was over a year old at SwRI's test track, and on maps on public roads that were several months old, but a more-rigorous study is warranted. In addition to characterizing the effect of surface degradation on localization performance, it would be beneficial to build a self-healing capability into the maps. In nearly all cases of surface modification through degradation and repair, short of complete repaving, self-healing would enable maps to always be up to date.

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